OTTO VON GUERICKE UNIVERSITÄT MAGDEBURG | NAT | FAKULTÄT FÜR NATURWISSENSCHAFTEN

Materials Science: Team Project

Subject:

Stacking faults in GaN

Description:

Stacking faults are structural defects in the crystal lattice that occur when the normal stacking sequence is interrupted. These defects can act as two-dimensional quantum wells in GaN, leading to localized states and characteristic photoluminescence (PL) emission, which can be used to study their properties. Stacking faults are often detrimental to device performance.

Work packages:

- **Literature Review**: The main focus of the project is conducting a comprehensive literature review on the chosen topic. Students should gather relevant papers and research articles.
- **Problem Identification**: Identify key problems or challenges within the chosen topic. This should involve understanding the current state of research and the unresolved issues in the field.
- **Methodological Approaches**: Students should investigate various approaches and methodologies used in the literature to address the identified problems.
- **Comparison of Approaches**: Compare and contrast the different methods found in the literature, discussing their strengths, weaknesses, and applicability to the identified problems.
- **Project Motivation**: At the end of the project, students should write a one-page summary introducing the project, explaining the motivation behind the topic and its relevance.
- **Visual Elements**: Students may include images or other visual materials to support their motivation page.
- **Presentation**: Prepare a presentation summarizing the findings and progress made during the project. This should include insights from the literature research, proposed models, and solutions explored.
- **Discussion**: After the presentation, a discussion should follow where students can explain their work, address questions, and evaluate the effectiveness of the approaches explored.